

CS3001 – Computer Networks
Quiz – 1B

Name	
Roll. No.	
Section	

Part 1 (3 Marks)

Provide 1-line answers to the following questions:

Why is the layered service-model important for scalability, flexibility and evolution of the layers?

Layering lets each protocol evolve independently, enabling flexible scaling without disrupting other layers.

Why is access network design important for overall Internet performance?

Access networks connect edge devices to the core, influencing bandwidth availability and latency

In a packet-switched network, what does throughput represent?

The actual data rate achieved (after accounting for protocol overhead and losses).

Part 2 (7 Marks)

Scenario

- 2 intermediate nodes, each adding **50 ms** of processing per packet.
- Packet size: **2048 bytes**.
- File to transfer: **20 000 000 bytes**.
- Minimum bandwidth of the link: **0.5 Mbps**.

How many seconds will it take to transfer the file from server to client?

Solution

STEP	CALCULATION	RESULT
Number of packets	$N = \lceil \frac{20\,000\,000}{2048} \rceil = 9766$	9766 packets
Transmission time per packet	$T_{tx} = \frac{16\,384}{0.5 \times 10^6} = 0.032768s = 32.768\text{ ms}$	32.768ms
Processing time per packet (per node)	$t_{proc} = 50\text{ ms}$	50ms
Bottleneck?	$t_{proc}(50\text{ ms}) > T_{tx} \rightarrow$ processing is the bottleneck.	–
Pipeline throughput	1 packet every 50 ms.	–
Time for first packet to finish	$T_{first} = T_{tx} + 2 \times t_{proc} = 32.768\text{ ms} + 100\text{ ms} = 132.768\text{ ms}$	132.768ms
Total transfer time	$T_{total} = T_{first} + (N - 1) \times t_{proc}$	
	$= 132.768\text{ ms} + 9765 \times 50\text{ ms}$	
	$= 132.768 + 488\,250\text{ ms} = 488\,382.768\text{ ms}$	
	$= 488.38\text{ s } (\approx 8\text{ min } 8\text{ s})$	488.4 s